

Reproduction Report: G2 — Guided Generation for Enhanced Output Diversity in LLMs

Overview

This report documents the reproduction of the main results from **G2: Guided Generation for Enhanced Output Diversity in LLMs** (EMNLP 2025). G2 is a training-free, plug-and-play decoding method that improves output diversity by combining a base model with a positive diversity guide and a negative consistency anchor via entropy-weighted logit arithmetic.

Environment

Item	Details
Model	meta-llama/Meta-Llama-3-8B-Instruct
Hardware	NVIDIA A100 80GB
Inference env	g2 — Python 3.10, transformers==4.41.0, vllm==0.5.0.post1
Evaluation env	g2_eval — Python 3.10, transformers==4.45.1, sentence-transformers==4.0.2
Key hyperparameters	= 0.3, temperature = 1.0

Benchmark 1: NoveltyBench Open-Ended Generation

Setup: 20 prompts from the curated split of yimingzhang/novelty-bench (creative writing, randomness, factual knowledge, subjective opinion). G2 generates 10 diverse responses per prompt. Semantic distinctness is evaluated with an NLI-based equivalence classifier (`partition.py`); quality is scored with Skywork-Reward-Gemma-2-27B-v0.2.

NoveltyBench Score

Metric	Reproduced	Paper	Diff
Mean Distinct (↑ more diverse)	6.01 / 10	5.80 / 10	+0.21
Mean Reward Score (↑ higher quality)	7.681 / 10	7.790 / 10	−0.109

Diversity Metrics (across all 10 outputs per input)

Metric	Reproduced	Paper	Diff
EAD ($\uparrow$ more diverse)	0.7677	0.7801	−0.0124
Sentence-BERT distance ($\uparrow$ more diverse)	0.3644	0.3791	−0.0147
Self-BLEU QS ($1 - \text{self-BLEU}$, $\uparrow$ more diverse)	62.75%	64.72%	−1.97%
Div Score (EAD + Div-BLEU)/4 + SentBERT/2	0.5310	0.5464	−0.0154

Benchmark 2: WMT’14 DE→EN Translation

Setup: 3,003 test sentences from WMT’14 (`wmt14`, `de-en`). G2 generates 5 diverse translations per sentence (iter 0 = greedy baseline, iter 1–4 = G2-guided). Quality measured with BLEU and COMET; diversity measured with EAD, Sentence-BERT distance, and Self-BLEU.

Quality Metrics (per iteration)

Reported metric: **Avg Quality** = (BLEU + COMET) / 2

Iteration	BLEU	COMET	Avg Quality
0 (baseline, greedy)	0.2783	0.8510	0.5647
1	0.2383	0.8352	0.5368
2	0.2290	0.8344	0.5317
3	0.2300	0.8333	0.5317
4	0.2292	0.8353	0.5323
Avg (iter 0–4)	—	—	0.5394
Avg (iter 1–4)	—	—	0.5331

Diversity Metrics (across all 5 outputs per input)

Metric	Mean	Std
EAD ($\uparrow$ more diverse)	0.5650	± 0.1332
Sentence-BERT distance ($\uparrow$ more diverse)	0.1104	± 0.0795
Self-BLEU QS ($1 - \text{self-BLEU}$, $\uparrow$ more diverse)	33.17%	—
Div Score (EAD + Div-BLEU)/4 + SentBERT/2	0.2794	—

Comparison with Paper (=0.3, est. from Figure)

Quality Score = Avg Quality $\times$ 100; Diversity Score = Div Score $\times$ 100

Metric	Paper (est.)	Reproduced	Diff
Quality Score	~53.8	53.31	-0.49
Diversity Score	~27.6	27.94	+0.34

Benchmark 3: XLSum Multilingual Summarization

Setup: 1,000 test examples from HuggingFace (GEM/xlsum). G2 generates 5 diverse summaries per article. Quality measured with ROUGE-1, ROUGE-2, and ROUGE-L (averaged); diversity measured with EAD, Sentence-BERT distance, and Self-BLEU.

Quality Metrics (per iteration)

Reported metric: **Avg Quality = (ROUGE-1 + ROUGE-2 + ROUGE-L) / 3**

Note: The current codebase (`eval/xlsum/run_eval.py`) only computes ROUGE-L. ROUGE-1 and ROUGE-2 were not saved in the existing results and cannot be recovered without re-running generation.

Iteration	ROUGE-L	Avg Quality
0 (baseline, greedy)	0.1530	N/A
1	0.1563	N/A
2	0.1585	N/A
3	0.1559	N/A
4	0.1570	N/A

Diversity Metrics (across all 5 outputs per input)

Metric	Mean	Std
EAD ($\uparrow$ more diverse)	0.7441	± 0.0583
Sentence-BERT distance ($\uparrow$ more diverse)	0.2036	± 0.0542
Self-BLEU QS ($1 - \text{self-BLEU}$, $\uparrow$ more diverse)	58.11%	—
Div Score (EAD + Div-BLEU)/4 + SentBERT/2	0.4331	—

Comparison with Paper (=0.3, est. from Figure)

Quality Score = Avg Quality $\times$ 100; Diversity Score = Div Score $\times$ 100

Metric	Paper (est.)	Reproduced	Diff
Quality Score	~14.6	N/A	—
Diversity Score	~43.0	43.31	+0.31

Observations

1. **Quality preservation:** On WMT, COMET drops modestly from 0.851 (iter 0) to ~0.833–0.835 for G2 iterations — a controlled quality–diversity trade-off rather than a collapse. On XLSum, COMET remains stable across iterations (~0.673–0.677).
2. **Increasing diversity across benchmarks:** EAD and Sentence-BERT distance scores are notably higher for XLSum (0.744, 0.204) and Novelty-Bench (0.768, 0.364) than for WMT (0.565, 0.110), reflecting the greater structural flexibility in open-ended generation vs. constrained translation.
3. **Entropy-weighted intervention:** The method’s selective intervention (only at high-entropy token positions) appears to explain the quality preservation — the model retains confident token predictions while nudging diversity at ambiguous positions.
4. **Context compression (i 3):** The embedding-based diverse-context selection helps prevent prompt length explosion in the 10-iteration NoveltyBench setting while keeping context informative.

Reproduction Commands

```
# Generation (g2 env)
conda activate g2
bash scripts/eval/wmt/run_eval.sh
bash scripts/eval/xlsum/run_eval.sh
bash scripts/eval/novelty/run_eval.sh
```

```
# Diversity metrics (g2_eval env)
conda activate g2_eval
bash scripts/eval/wmt/diversity.sh
bash scripts/eval/xlsum/diversity.sh
bash scripts/eval/novelty/diversity.sh
```

All scripts run from the project root (emnlp25-g2/). Output files are written to results/{wmt,xlsum,novelty}/.